

## INTRODUCTION

**Background:**

- Sepsis associated encephalopathy (SAE) is a poorly understood acute cerebral dysfunction that frequently appears in the setting of sepsis induced systemic inflammation
- SAE may appear early in the course of sepsis, yet there is a lack of specific investigations available to clinicians to determine long-term patient outcomes
- Contemporary serum biomarkers have not yet had a role in the neuro-prognostication of the ED SAE patient.

**Objective:**

- We investigated the diagnostic and prognostic utility of neurofilament heavy chain (pNFH) and S100 calcium-binding protein B (S100B), markers of axonal and blood brain barrier integrity, to assess their role in predicting cognitive performance and outcomes in ED sepsis

## METHODS

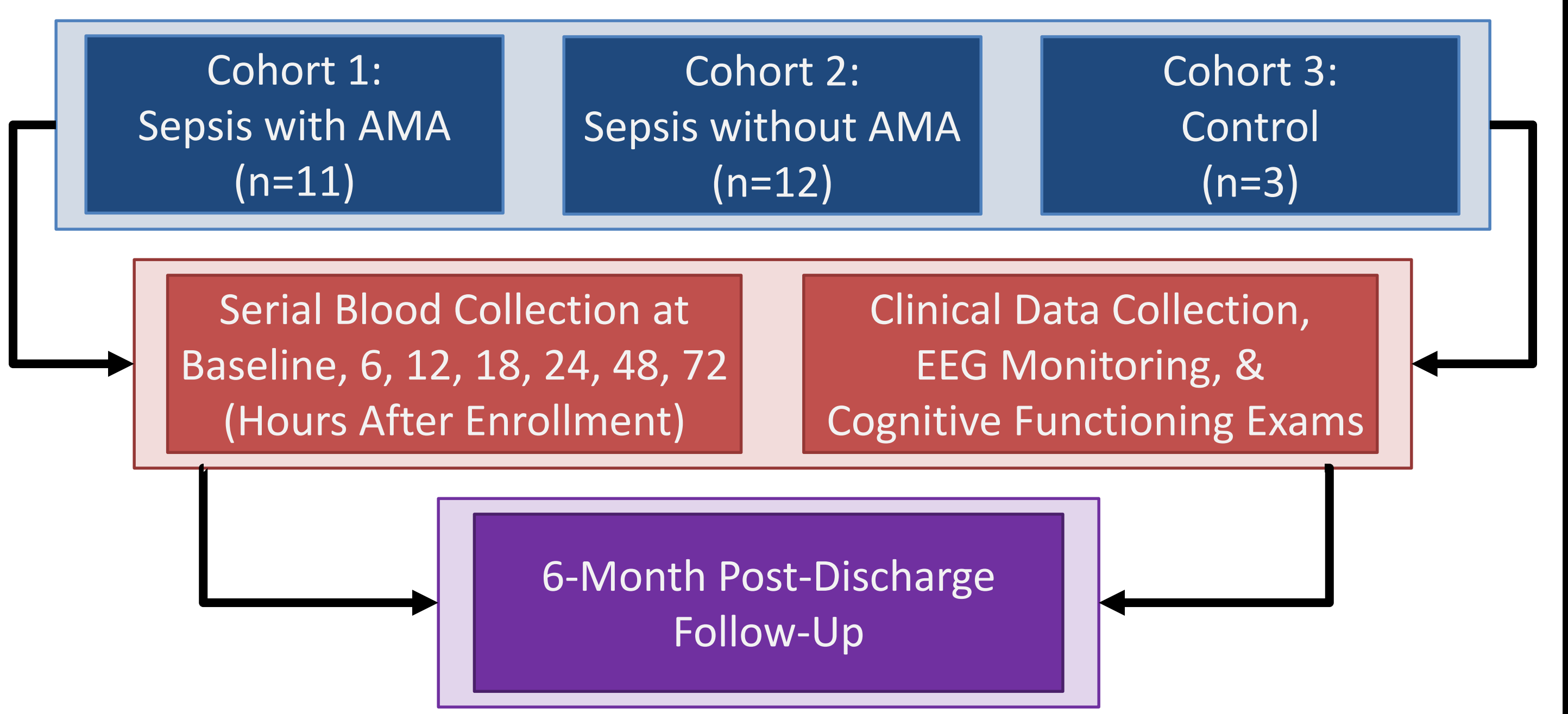
**Study Design**

- A single center, prospective, longitudinal study of ED patients >18 years with suspected or confirmed infection and systemic inflammatory response syndrome (SIRS) criteria was performed (IRB: 201700135)
- Patients with past medical history of neurodegenerative disease, acute brain injury, renal failure, liver failure, or severe anemia were excluded.
- Subjects were recruited into one of three cohorts: (1) Sepsis with AMS (2) Sepsis without AMS (3) Control, No Systemic Inflammation
- Determination of AMS at enrollment was defined a priori by treating clinician
- Serial serum collection for biomarker analysis was performed 0, 6, 12, 18, 48, and 72 hours after enrollment.
- Outcomes included cerebral performance category (CPC) and mortality at discharge and at 6-months.

**Statistical Methods**

- Associations between sepsis patients with and without AMS and outcomes were tested by Fisher’s exact test and Mann-Whitney U-test. (Table 1 & 2)
- Utility of serum biomarkers in predicting cohort assignment and outcomes was assessed using receiver operator characteristic (ROC) analysis. Optimal cutoff points were identify by Youden’s J statistic (Table 3)
- Generalized estimating equations (GEE) models, adjusted for within-subject correlation, analyzed relations in biomarker kinetics and outcome (Figure 2 & 3)

Figure 1: Flowchart of Study Procedures



## RESULTS

**Table 1: Demographics of study population by cohort at enrollment (n=26)**  
\*Includes Cohort 3 subjects

|                             | Total*<br>(n=26) | Cohort 1<br>(n=11) | Cohort 2<br>(n=12) | P value |
|-----------------------------|------------------|--------------------|--------------------|---------|
| Age – yrs                   | 60.3 ± 19.1      | 70.3 ± 18.3        | 51.7 ± 18.4        | 0.032   |
| Sex – no. (%)               |                  |                    |                    |         |
| Female                      | 10 (38.5)        | 4 (36.4)           | 5 (41.7)           | 0.999   |
| Male                        | 16 (61.5)        | 7 (63.6)           | 7 (58.3)           |         |
| Racial Group – no. (%)      |                  |                    |                    |         |
| African American            | 6 (23.1)         | 1 (9.1)            | 4 (33.3)           | 0.317   |
| White                       | 20 (76.9)        | 10 (90.9)          | 8 (66.7)           |         |
| Body Mass Index – Mean (SD) | 27.8 ± 8.7       | 25.3 ± 4.3         | 28.3 ± 10.1        | 0.381   |
| APACHE II Score (Baseline)  | 16.8 ± 6.3       | 19.5 ± 8.4         | 15.2 ± 3.5         | 0.156   |

**Table 3: Area Under ROC Curve and Cutpoints\* for (A) Cohort Assignment, (B) CPC Score at Discharge, and (C) 6-Month Mortality using S100B and pNFH**  
\*Presented as concertation value (pg/mL) or relative percent change (%). Cutpoints only provided for measures with significant AUC (p<0.05)

**(A) Cohort 1 Assignment (vs. Cohort 2)**

|                  | AUC (95CI)       | P Value | Cutpoint | Sens. (%) | Spec. (%) | PPV (%) | NPV (%) |
|------------------|------------------|---------|----------|-----------|-----------|---------|---------|
| <b>pNFH</b>      |                  |         |          |           |           |         |         |
| [Baseline]       | 0.73 (.55 – .97) | 0.016   | >168.9   | 90.9      | 58.3      | 66.7    | 87.5    |
| [H24 – Baseline] | 0.81 (.60 – .99) | 0.004   | < -1.0%  | 87.5      | 72.7      | 70.0    | 88.9    |
| [H72 – Baseline] | 0.89 (.66 – .99) | <0.001  | < -2.2%  | 83.3      | 100.0     | 100.0   | 85.7    |
| <b>S100B</b>     |                  |         |          |           |           |         |         |
| [H72]            | 0.87 (.64 – .99) | 0.002   | < 32.2   | 80.0      | 83.3      | 80.0    | 83.3    |
| [H72 – Baseline] | 0.80 (.51 – .99) | 0.048   | < -73.6% | 60.0      | 100.0     | 100.0   | 75.0    |

**(B) CPC Score at Discharge 1-2 (vs. Score 3-5)**

|                  | AUC (95CI)       | P Value | Cutpoint | Sens. (%) | Spec. (%) | PPV (%) | NPV (%) |
|------------------|------------------|---------|----------|-----------|-----------|---------|---------|
| <b>pNFH</b>      |                  |         |          |           |           |         |         |
| [Baseline]       | 0.77 (.55 – .99) | 0.018   | < 211.9  | 80.0      | 71.4      | 40.0    | 93.8    |
| [H48]            | 0.87 (.58 – .99) | 0.012   | < 209.1  | 100.0     | 73.3      | 33.3    | 100.0   |
| [H72]            | 0.91 (.71 – .99) | <0.001  | < 280.3  | 100.0     | 72.7      | 40.0    | 100.0   |
| [H6 – Baseline]  | 0.82 (.66 – .98) | <0.001  | > -2.1%  | 100.0     | 76.2      | 50.3    | 100.0   |
| [H18 – Baseline] | 0.77 (.57 – .97) | 0.007   | > -1.4%  | 100.0     | 61.9      | 33.2    | 100.0   |
| [H24 – Baseline] | 0.87 (.68 – .99) | <0.001  | > -2.6%  | 100.0     | 72.2      | 37.5    | 100.0   |
| <b>S100B</b>     |                  |         |          |           |           |         |         |
| [H72 – Baseline] | 0.91 (.69 – .99) | <0.001  | > -73.6% | 100.0     | 81.8      | 50.0    | 100.0   |
| [H72 – H48]      | 0.99 (.99 – .99) | <0.001  | > -79.7% | 100.0     | 100.0     | 100.0   | 100.0   |

**(C) Mortality Within 6-Months (vs. Survival)**

|                  | AUC (95CI)       | P Value | Cutpoint | Sens. (%) | Spec. (%) | PPV (%) | NPV (%) |
|------------------|------------------|---------|----------|-----------|-----------|---------|---------|
| <b>pNFH</b>      |                  |         |          |           |           |         |         |
| [Baseline]       | 0.75 (.50 – .99) | 0.052   | > 211.9  | 66.7      | 76.5      | 50.0    | 86.7    |
| [H48 – H24]      | 0.83 (.59 – .99) | 0.008   | < 1.3%   | 100.0     | 70.0      | 50.0    | 100.0   |
| <b>S100B</b>     |                  |         |          |           |           |         |         |
| [H48 – H24]      | 0.85 (.63 – .99) | 0.002   | < -7.3%  | 100.0     | 76.9      | 50.0    | 100.0   |
| [H72 – H48]      | 0.88 (.63 – .99) | 0.003   | < -78.6% | 100.0     | 87.5      | 50.0    | 100.0   |
| [H72 – Baseline] | 0.94 (.79 – .99) | <0.001  | < -73.6% | 100.0     | 88.9      | 66.7    | 100.0   |

**Table 2: Primary outcomes by cohort assignment at enrollment**  
\*Includes Cohort 3 subjects

|                              | Total*<br>(n=26) | Cohort 1<br>(n=11) | Cohort 2<br>(n=12) | P value |
|------------------------------|------------------|--------------------|--------------------|---------|
| <b>CPC Score Discharge</b>   |                  |                    |                    |         |
| 1 – 2 (Good)                 | 18 (78.3)        | 6 (54.5)           | 12 (100.0)         | 0.008   |
| 3 – 5 (Poor)                 | 5 (21.7)         | 5 (45.5)           | 0 (0.0)            |         |
| <b>Survival to Discharge</b> |                  |                    |                    |         |
| Yes                          | 24 (92.3)        | 9 (81.8)           | 12 (100.0)         | 0.217   |
| No                           | 2 (7.7)          | 2 (18.2)           | 0 (0.0)            |         |
| <b>Survival to 6-Months</b>  |                  |                    |                    |         |
| Yes                          | 17 (73.9)        | 5 (50.0)           | 9 (90.0)           | 0.141   |
| No                           | 6 (26.1)         | 5 (50.0)           | 1 (10.0)           |         |

Figure 2: Percent Change in [S100B] From Baseline at Each Study Time Point by 6-Month Mortality (p=0.007)

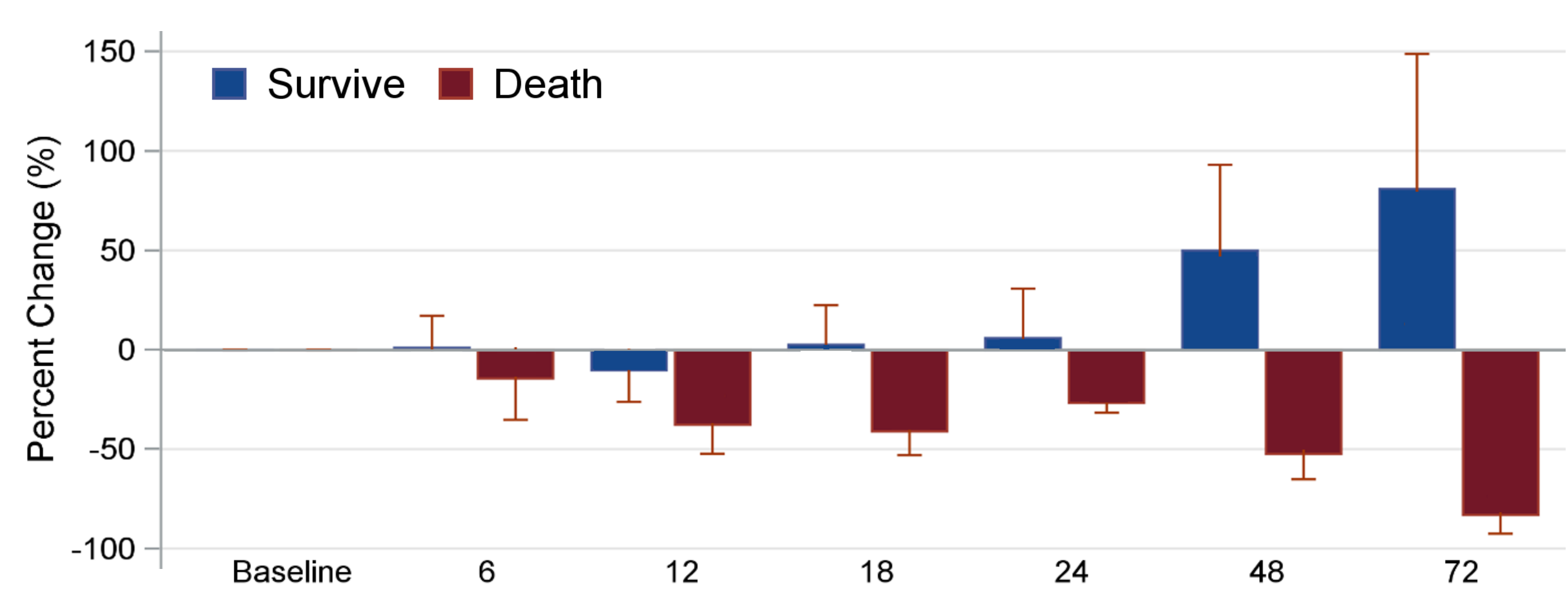
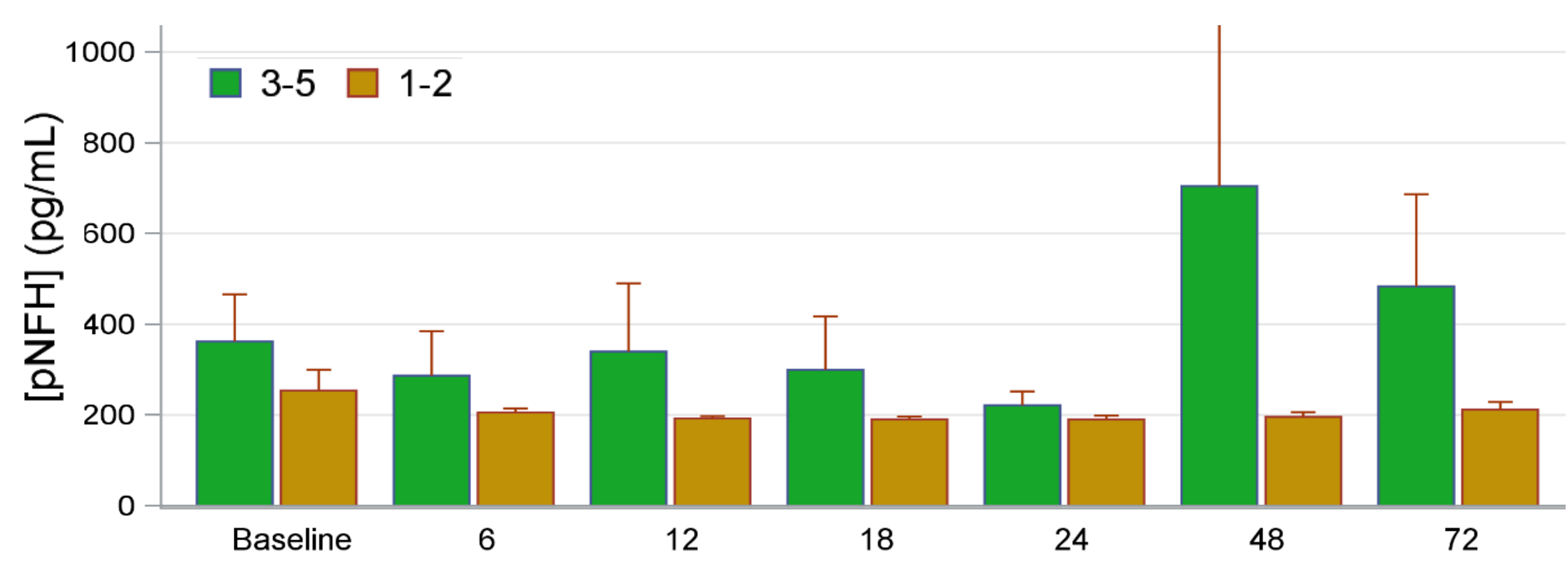


Figure 3: Mean [pNFH] at Each Study Time Point by CPC Score at Discharge (p<0.001)



## CONCLUSION

- In this pilot study, we confirm that the acute presentation of SAE is associated with cognitive dysfunction at hospital discharge.
- Novel biomarkers S100B and pNFH demonstrate discriminatory utility in the diagnosis of acute SAE, prognostication of persistent cognitive dysfunction and long-term survival among sepsis patients.
- Potential S100B and pNFH cutoff points have been identified for further diagnostic and prognostic testing.
- Future research in this cohort study will integrate new biomarkers, MRI results, EEG monitoring, and gut microbiota patterns in order to test their association with outcomes and identify the causal pathway of SAE